Maninder Singh

+91-9877945982 — captionmanindersingh@gmail.com — GitHub — LinkedIn

Education

Dr. B.R. Ambedkar National Institute of Technology (NIT), Jalandhar **Expected Graduation: 2028**
Bachelor of Technology in Information Technology **CGPA: 8.71**

Skills

- **Programming Languages:** Python, C, C++, JavaScript
- **Machine Learning:** PyTorch, Scikit-Learn, NumPy, Pandas, Matplotlib, Deep Learning, Transfer Learning
- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend:** Node.js, Express.js
- **Database:** MongoDB, MySQL
- **Tools:** Git, GitHub, Linux, VS Code, Kaggle
- **Core Concepts:** Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, Computer Networks, DBMS, Machine Learning

Projects

Land Cover Classification using ResNet50 Transfer Learning Kaggle

- Improved validation accuracy from **32% to 80%+** on the LUCAS land-use dataset.
- Applied transfer learning using **ResNet50** pretrained on ImageNet.
- Performed data augmentation using flips, rotations and color jitter.
- Addressed class imbalance using **WeightedRandomSampler**.
- Implemented learning-rate scheduling for stable model convergence.
- Built using PyTorch, Torchvision, Scikit-Learn and Kaggle Notebooks.

Diabetes Prediction using Machine Learning GitHub

- Developed a Random Forest classifier for diabetes prediction.
- Handled missing values using SimpleImputer.
- Performed preprocessing and feature engineering.
- Evaluated performance using Precision, Recall, F1-Score and Confusion Matrix.

Agri Sales Management System GitHub

- Developed a full-stack agricultural sales platform using the MERN stack.
- Designed responsive user interfaces for product listings and farmer interactions.
- Integrated frontend components with backend APIs.
- Used Git/GitHub for version control and collaborative development.

Achievements

- District-Level Under-17 Cricket Player
- Built multiple AI/ML and Full-Stack MERN projects
- Maintained a CGPA of 8.71 at NIT Jalandhar